# Design and Optimisation of Si/SiGe Punch-Through Diodes using TCAD Simulations

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## Device Structure and Methodology

**Punch-through diode**

A diode, in which at a certain voltage in reverse voltage depletion region spans through the device, leading to a direct conduction path and, hence, strong current.

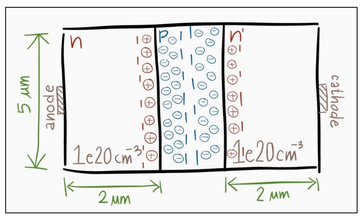



*Fig 1. Structure of the npn diode used for analysis*

**npn diode:**
- Length of diode: 5 um
- Thickness of n-regions: 2 um
- Doping of n-regions: $10^{20}$ $cm^{-3}$

**Sentaurus Specifications:**
- Constant doping – As and B
- Dense meshing in the p-region
- Approx. 7386 mesh grid points

**Methodology:**
- pn diode simulations to analyse a fundamental a pn diode on Sentaurus
- - npn diode with Si in all sections (n, p and n)
  - Replacing Si with SiGe in p-region
- Varying the p-region (base) dopings and lengths; started with larger widths and decreased further
- Analyzing the electrostatic profiles and the IV characteristics with change in parameters: material, length, dopings

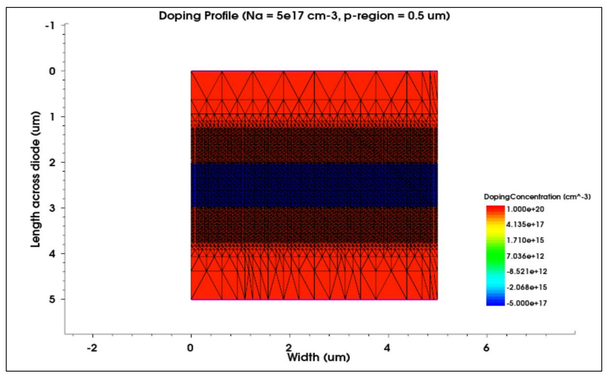
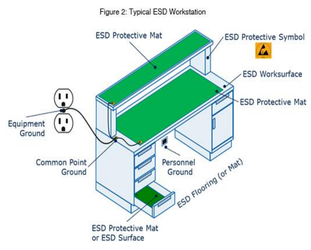


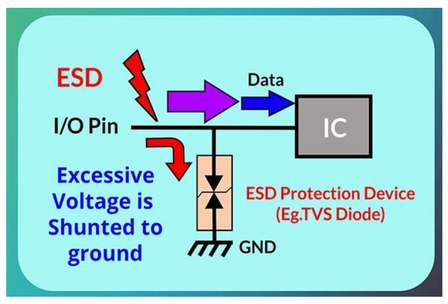

*Fig 2. Meshing profile of the diode in Sentaurus*

## Objective

- Study punch-through diode behavior using TCAD simulations.
- Evaluate effects of material (Si/SiGe) and device parameters.
- Optimize design for improved ESD protection performance.

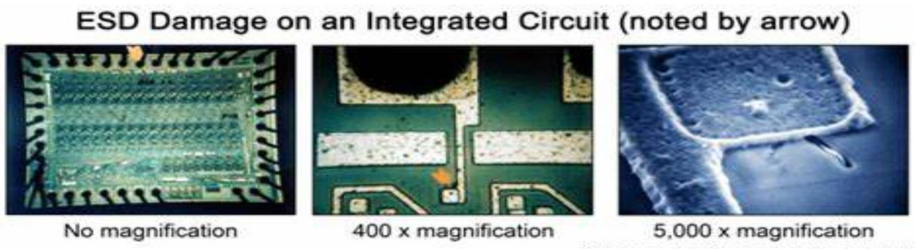



## Working of ESD device




- An ESD event creates a high-voltage spike at the I/O pin that can damage the IC.
- The protection device provides a low-resistance path, diverting the excess voltage.
- The surge is safely discharged to ground, protecting the circuit.

## Failure due to ESD

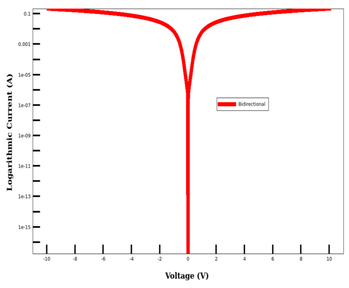



**Mechanism**
- High-voltage spike creates intense electric field across device regions.
- Thin gate oxide breaks down due to dielectric failure.
- Sudden current causes localized heating (Joule heating).
- Results in metal melting and junction damage.
- Leads to permanent degradation or device failure.

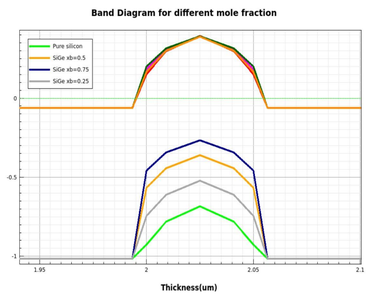
**Causes of ESD**
- Friction between materials
- Electrostatic induction from nearby charged objects
- Low humidity and dry environmental conditions
- Use of insulating materials (plastic, rubber, synthetic fabrics)

## Motivation for npn punch through diode

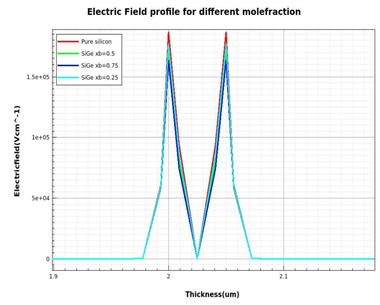
- Conventional ESD protection devices often face limitations in speed, scalability, and controlled conduction at advanced technology nodes.
- Punch-through diodes enable controlled conduction at a defined voltage by engineering depletion regions, making them suitable for fast transient protection.
- Incorporating SiGe material and optimized geometry offers improved electric field control and enhanced device performance.
- There is a need for robust, low-voltage, and reliable ESD protection solutions for modern electronics such as brain–machine interfaces.



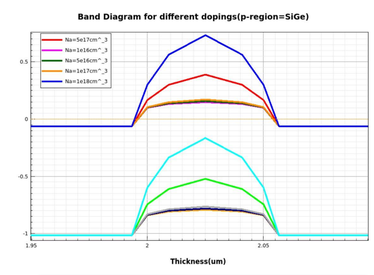

## Simulations Result at Equilibrium



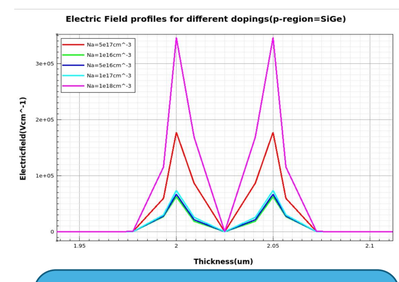

- Bandgap Narrowing: As more Germanium is added valence band significantly rises, decreasing the bandgap
- Most of the band offsets occur in the valence band




- Changing the mole fraction has relatively minor effect on the peak electric field
- The decrease occurs because adding Germanium increases the dielectric of the material




- High doping show the most significant spikes, indicating a high built in potential and larger barriers for carrier transport.
- Low doping curves show much flatter indicating a lower potential and more uniform electric field




- The electric field peaks at interfaces due to depletion regions from material transitions. Higher doping increases peak strength because of more fixed charges. Between interfaces, the field drops near zero, indicating charge neutrality and weaker internal electric fields.

*Table 1.*

**Electric Field vs Doping Concentration (Si vs $Si_{0.75}Ge_{0.25}$ in p-region)**

| Doping Concentration ($cm^{-3}$) | Pure Si Peak Electric Field ($\times 10^5$ V/cm) | $Si_{0.75}Ge_{0.25}$ (p-region) Peak Electric Field ($\times 10^5$ V/cm) |
|---|---|---|
| $1 \times 10^{16}$ | $7.30 \times 10^4$ | $6.24 \times 10^4$ ↓ 14.5% |
| $5 \times 10^{16}$ | $8.32 \times 10^4$ | $7.48 \times 10^4$ ↓ 10.1% |
| $1 \times 10^{17}$ | $1.16 \times 10^5$ | $1.14 \times 10^5$ ↓ 1.7% |
| $5 \times 10^{17}$ | $3.22 \times 10^5$ | $2.90 \times 10^5$ ↓ 9.9% |
| $1 \times 10^{18}$ | $4.22 \times 10^5$ | $3.75 \times 10^5$ ↓ 11.1% |

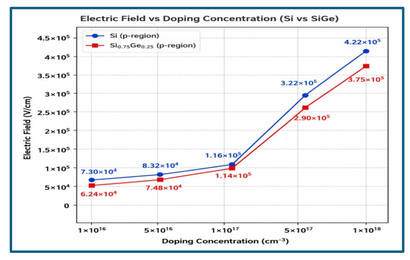



*Table 2.*

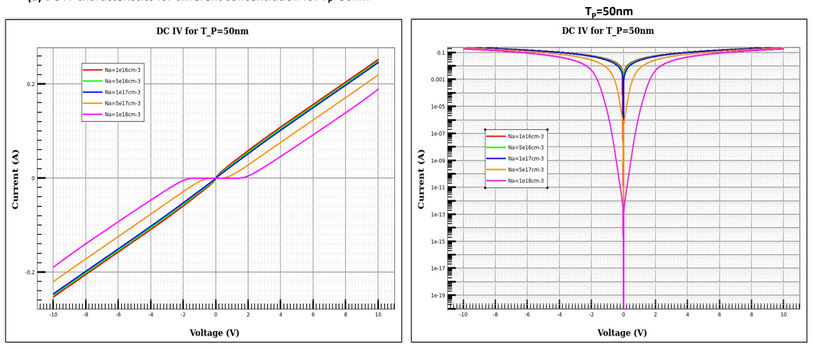
**N-type Concentration = $1 \times 10^{20}$ $cm^{-3}$**
**P-type Concentration = $5 \times 10^{17}$ $cm^{-3}$**
**P-region Thickness = 50 nm**

| Ge Mole Fraction ($x_b$) | Bandgap, $E_g$ (eV) | Peak Electric Field ($\times 10^5$ V/cm) |
|---|---|---|
| 0 (Pure Si) | 1.12 | 4.10 |
| 0.25 ($Si_{0.75}Ge_{0.25}$) | 1.03 | 3.75 |
| 0.50 ($Si_{0.5}Ge_{0.5}$) | 0.93 | 3.25 |
| 0.75 ($Si_{0.25}Ge_{0.75}$) | 0.80 | 2.85 |

## DC Analysis through the Simulation

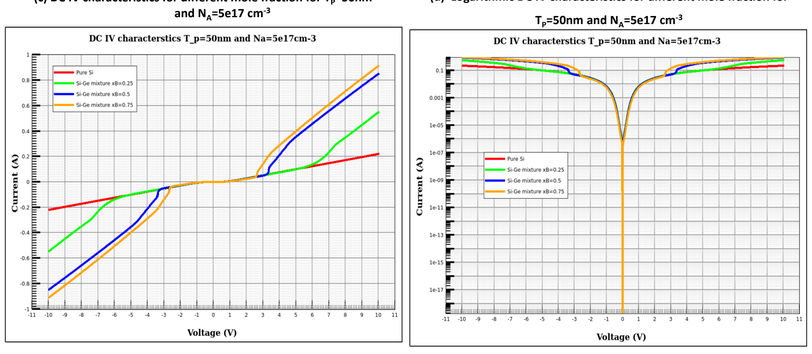
(a) DC IV characteristics for different concentration for $T_p$=50nm

(b) Logarithmic DC IV characteristics for different concentration for $T_p$=50nm




(c) DC IV characteristics for different mole fraction for $T_p$=50nm and $N_A$=5e17 $cm^{-3}$

(d) Logarithmic DC IV characteristics for different mole fraction for $T_p$=50nm and $N_A$=5e17 $cm^{-3}$




- Current increases linearly with voltage after conduction begins, indicating stable device behavior.
- As p-region doping increases, the turn-on voltage shifts higher, requiring more voltage for conduction.
- Lower doping shows earlier conduction, while higher doping delays the punch-through effect.
- At high doping levels, the device exhibits reduced current for the same voltage, indicating stronger control over carrier flow.

- The diode exhibits a distinct turn-on (punch-through) behavior, with current rising sharply after a threshold voltage.
- Pure silicon shows the lowest current, indicating higher resistance and delayed conduction.
- Incorporating SiGe increases current flow, demonstrating improved carrier transport compared to pure Si.
- Higher Ge content (x = 0.5 and 0.75) results in earlier and stronger conduction, due to reduced bandgap.

*Table 3.*

| Conduction Voltage vs. P region Thickness for Nd=1e20cm-3 and Na=5e17cm-3 | | | | |
|---|---|---|---|---|
| P-Region Thickness (nm) | Pure Si (V) | SiGe (x = 0.25) (V) | SiGe (x = 0.5) (V) | SiGe (x = 0.75) (V) |
| 50 | 5.1 | 4.4 | 2.2 | 1.5 |
| 70 | 5.5 | 4.8 | 2.7 | 1.9 |
| 90 | 5.9 | 5.1 | 3.2 | 2.3 |
| 110 | 8.2 | 6.2 | 3.8 | 2.8 |
| 130 | >10 | 6.6 | 4.0 | 3.7 |

*Table 4.*

| Conduction Voltage vs. P region concentration for T_p=50nm | | | | |
|---|---|---|---|---|
| P Doping ($cm^{-3}$) | Pure Si (V) | SiGe (x = 0.25) (V) | SiGe (x = 0.5) (V) | SiGe (x = 0.75) (V) |
| $1 \times 10^{16}$ | 0 | 0 | 0 | 0 |
| $5 \times 10^{16}$ | 0 | 0 | 0 | 0 |
| $1 \times 10^{17}$ | 1.8 | 1.3 | 1.0 | 0.8 |
| $5 \times 10^{17}$ | 5.1 V | 4.4 V | 2.2 V | 1.5 V |
| $1 \times 10^{18}$ | 5.6 V | 5.0 V | 2.7 V | 2.0 V |

## Conclusion & Future Work

### Conclusion
- Studied punch-through diode behavior and feasibility using TCAD simulations.
- Compared conventional Si structures with SiGe modified designs.
- Found that using SiGe in the p-region and tuning doping affects electric field, depletion width, and current conduction.
- Punch-through mechanism creates a controlled conduction path at higher bias, improving speed and performance.
- Structural optimization (reduced base width, doping) enhances carrier transport and controls electric field peaks.
- Concluded that device performance can be tuned effectively through material choice and geometry.

### Future Work
- Extend study to applications in ESD protection and low-frequency, high-voltage power electronics.
- Evaluate device under ESD stress for robustness, failure limits, and transient response.
- Optimize key parameters: holding voltage, trigger voltage, and current handling capability.
- Perform experimental validation and compare with standard ESD protection devices

## Acknowledgements

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